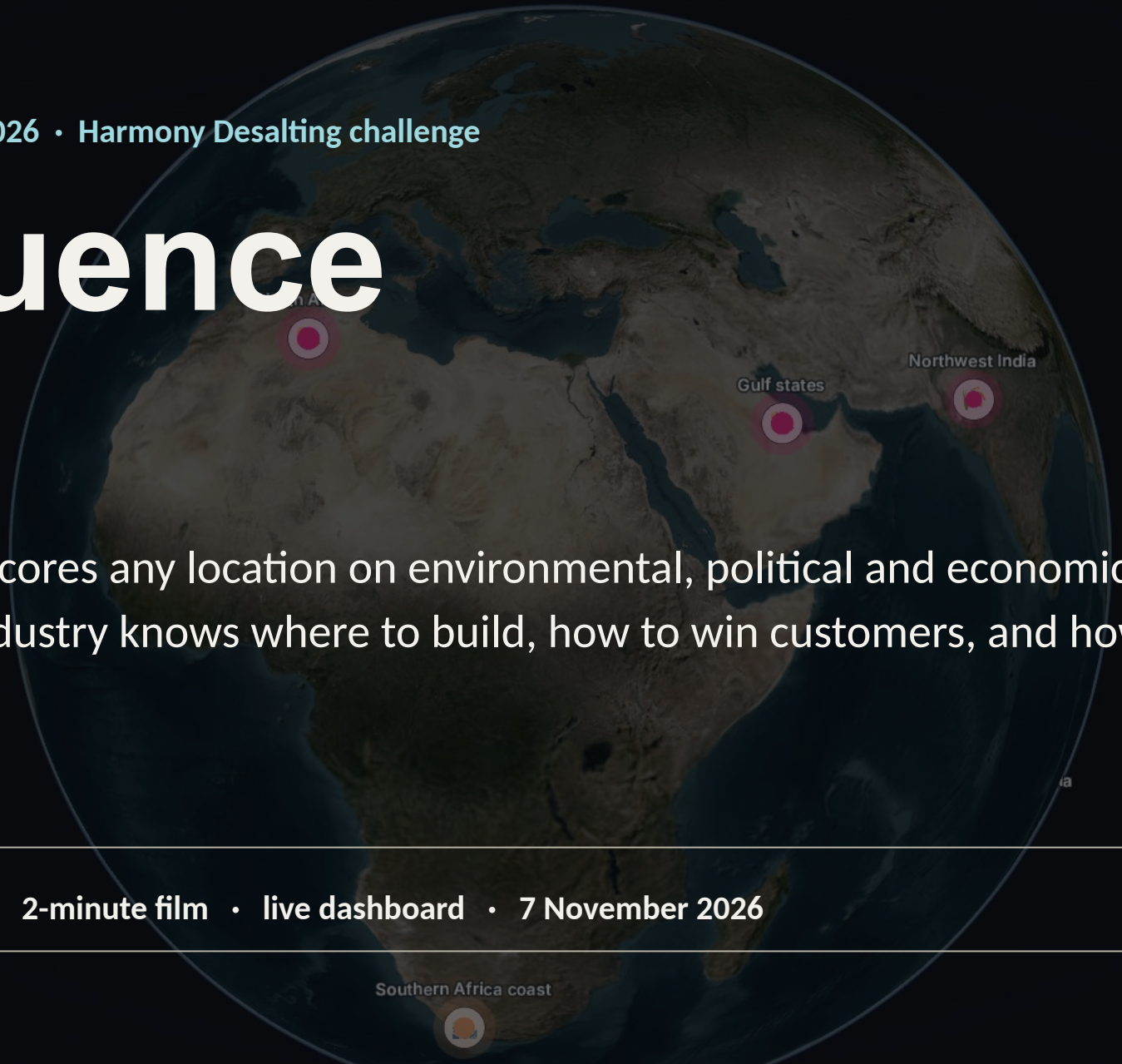
MIT Energy & Climate Hack 2026 · Harmony Desalting challenge

Confluence



One dashboard that scores any location on environmental, political and economic uncertainty, so the industry knows where to build, how to win customers, and how to reduce risk.

Team Sea Major · 9 slides · 2-minute film · live dashboard · 7 November 2026

Harmony Desalting: “What actions should the [desalination] industry take to secure our freshwater supplies in 2050 and beyond?”

The challenge, in Harmony's words

Reduce the uncertainty before you build.

"Setting aside technology development, what actions should the desalination industry take to secure our freshwater supplies in 2050 and beyond?"

Quantum Wei, Harmony Desalting

We read the problems behind that question across dozens of sources. Plants fail on uncertainty, not on technology. Three kinds decide whether the right plant gets built on time:

Environmental conditions

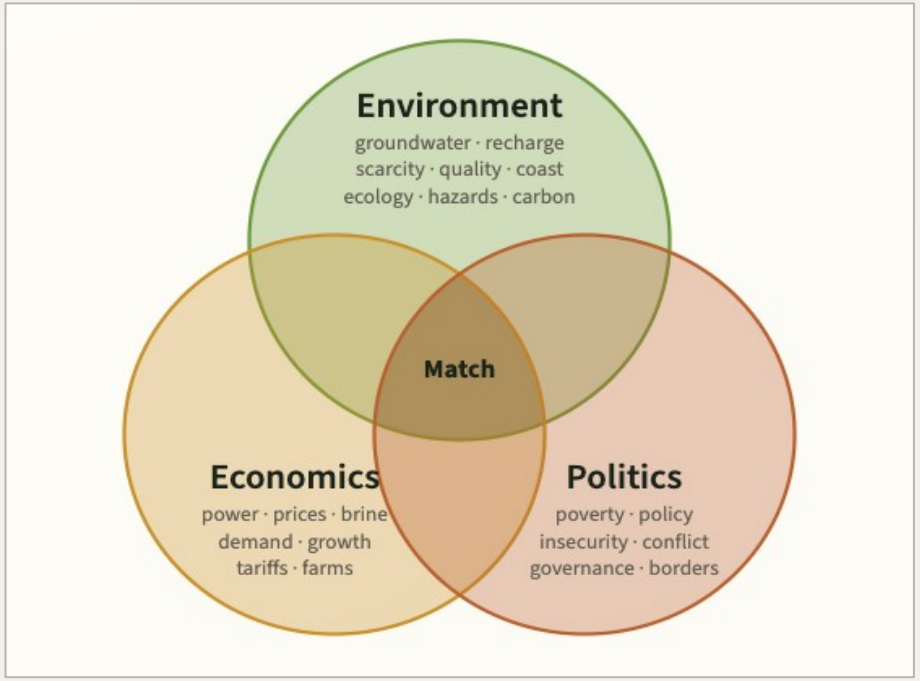
scarcity, groundwater, water quality, coast, ecology, hazards, clean energy

Political and governance conditions

governance, stability, shared rivers

Economic viability

ability to pay, energy cost, farm share, offtake

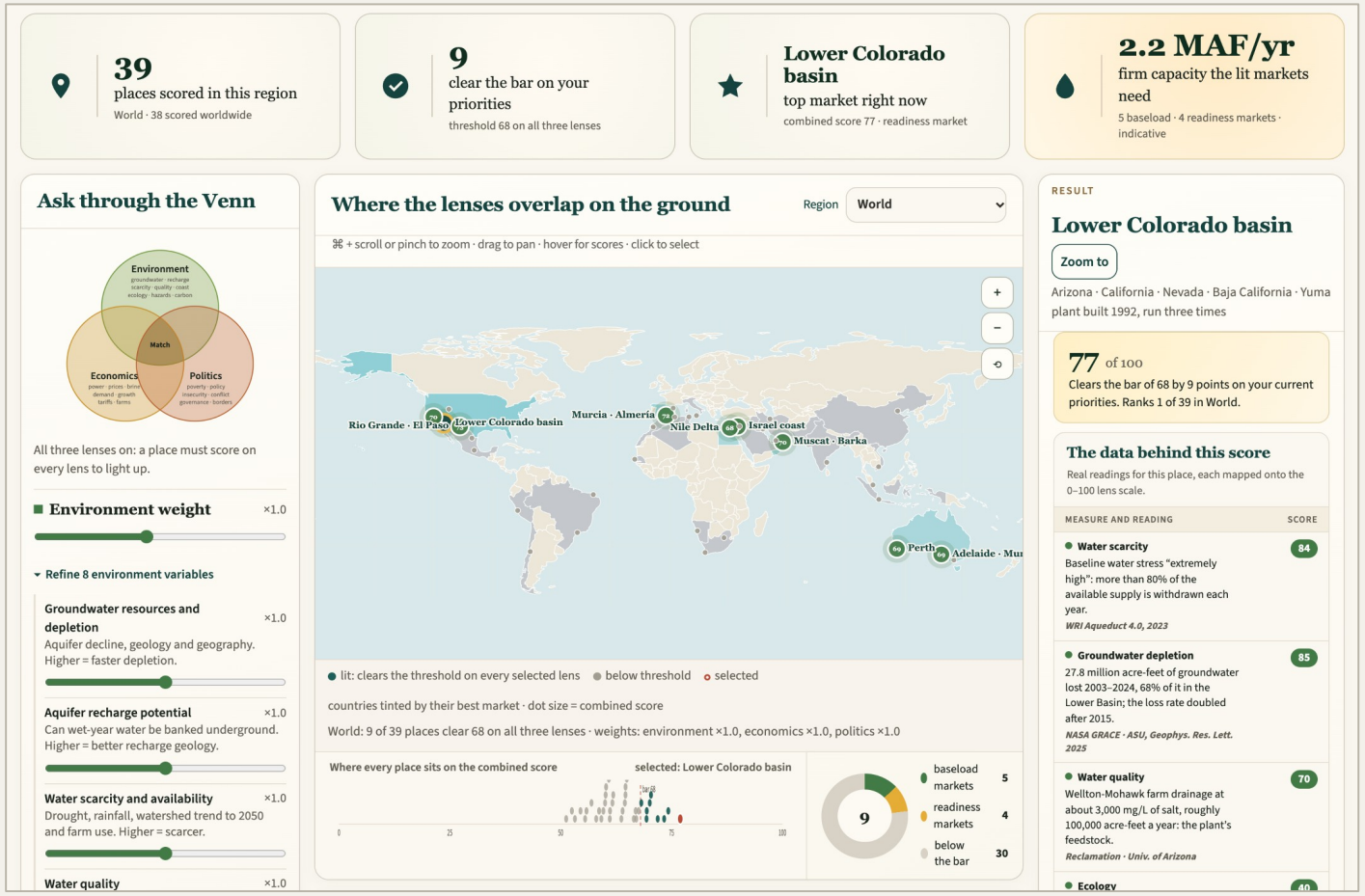


Where all three overlap is where a plant gets built, bought and paid for.

Three questions for the industry: where to build · how to acquire customers · how to reduce uncertainty

Our proposal: an adaptable model

Set your priorities. The model adapts.



1 Choose what matters

Weight the three lenses and your confidence in each variable. A social-impact fund and a desal company get different maps.

2 It scores every location

From public indices that already quantify these risks: WRI Aqueduct, WWF Water Risk Filter, Water Peace and Security, UN SDG 6, FAO AQUASTAT, NASA GRACE.

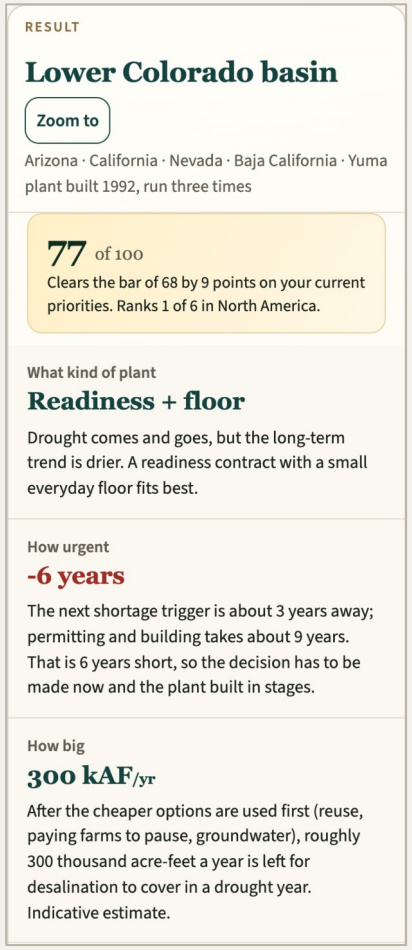
3 It ranks your region

Pick a region of interest and get the locations that fit your priorities, as investment opportunities with a score.

Prototype: the scoring methods exist (the WWF Water Risk Filter is the template); the full data back end is not built yet.

Case study: the tool applied to Yuma, Arizona

Set the toggles for a typical plant. Read the output.


RESULT

Lower Colorado basin

Zoom to

Arizona · California · Nevada · Baja California · Yuma plant built 1992, run three times

77 of 100

Clears the bar of 68 by 9 points on your current priorities. Ranks 1 of 6 in North America.

What kind of plant

Readiness + floor

Drought comes and goes, but the long-term trend is drier. A readiness contract with a small everyday floor fits best.

How urgent

-6 years

The next shortage trigger is about 3 years away; permitting and building takes about 9 years. That is 6 years short, so the decision has to be made now and the plant built in stages.

How big

300 kAF/yr

After the cheaper options are used first (reuse, paying farms to pause, groundwater), roughly 300 thousand acre-feet a year is left for desalination to cover in a drought year. Indicative estimate.

Readiness

mode: backup with a baseload floor

−6 years

timing gap: trigger in ~3, build takes ~9

300 kAF/yr

the right size, after cheaper water

The business pitch it returns

Sell to the binational commission and Reclamation, as treaty compliance.

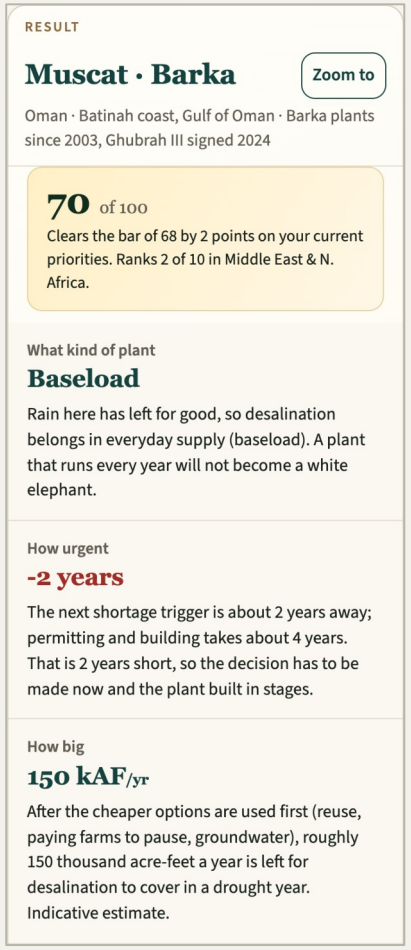
Water made at Yuma is water not taken from the shared river. Readiness payments, modules on a Lake Mead trigger, then an exchange with the farms.

Scored from real data · combined 77 of 100, first in North America

Measure	Reading and source	Score
Water stress	Aqueduct 4.0: extremely high, more than 80% of the supply withdrawn each year	84
Groundwater	GRACE: 27.8 million acre-feet lost 2003–24, 68% of it in the Lower Basin	85
Farm use	74% of direct human use is irrigation; alfalfa and hay alone take 46% (Richter 2024)	85
Shared river	1944 Treaty: 1.5 MAF a year to Mexico; Minute 242 salinity limit; Minute 323 desal	90
Shortage trigger	Lake Mead 1,056 ft: Tier 1 shortage in 2026, below 1,050 ft by spring 2027	≈3 yrs
Plant record	Built 1992 for $250M; ran 1992, 2007 (10%), 2010–11 (one-third capacity, 30,000 AF)	readiness

Case study 2: Muscat · Barka, Oman

Same toggles, run on the Middle East.



Baseload

mode: everyday supply, rain has left for good

−2 years

timing gap: deficit now, build takes ~4

150 kAF/yr

the right size, after cheaper water

The business pitch it returns

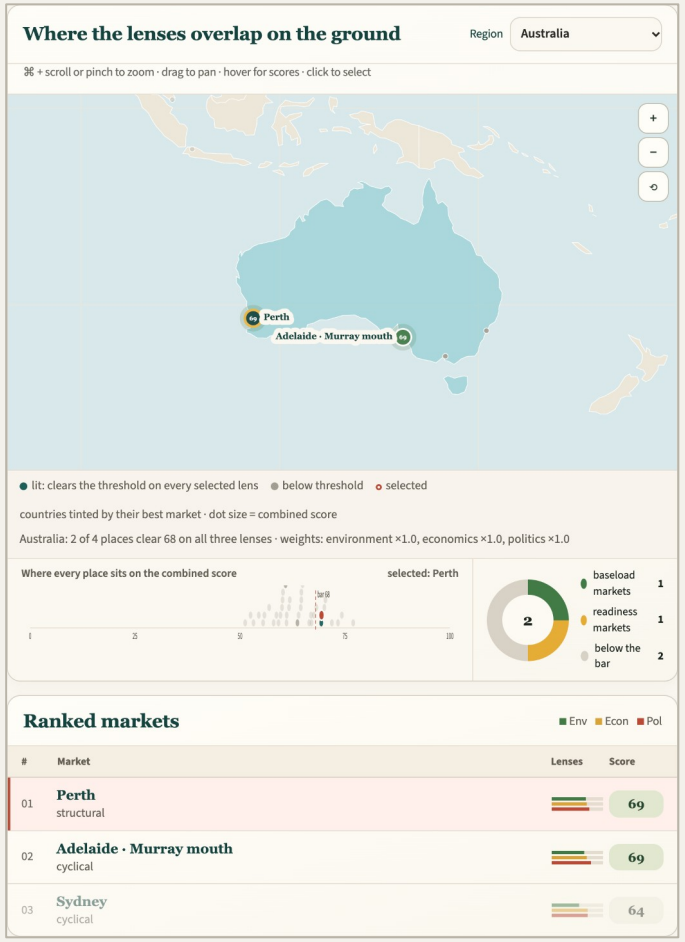
Sell to Nama PWP, the state's single buyer, as everyday supply on a long contract.

Price it against the next-cheapest source, the failing coastal wells, not rain. Pair it with an in-lieu exchange: desal for Muscat keeps farm pumps off the Batinah aquifer, and 174 recharge dams bank the wet years.

Scored from real data · combined 70 of 100, second in the region behind Israel's finished market

Measure	Reading and source	Score
Water scarcity	≈300 m^3 per person a year, under the 500 m^3 scarcity line; 2024 deficit ≈213 million m^3	92
Groundwater	Batinah water table down 5 m; coastal farms lost to salt; farms pump 93% of groundwater	85
Farm use	Agriculture and green areas take about 85% of national water demand	85
Ability to pay	Investment grade with S&P (2024), Moody's and Fitch (2025); government debt ≈36% of GDP	80
Policy	One buyer, Nama PWP: Ghubrah III ($325M) signed 2024; Barka V in service since Aug 2024	85
Shared river	None: no treaty lever, but no upstream dam risk either	25

It returns the modes that worked.



	Tool says	What happened
Perth	**Everyday supply (baseload)**	Built 2006, before the crisis. Runs every year.
Melbourne	**Readiness contract**	Bought in a panic in 2007. Finished as the dams refilled. Idle five years.
Sydney	**Readiness contract**	Switched off in 2012, restarted on a rule in 2019.
Adelaide	**Readiness, then exchange**	Paid to run in 2019 so farmers upstream kept river water.

Same weights for all four. The one that failed, Melbourne, failed on timing, which is exactly the gap the tool reports.

Every way to get fresh water, cheapest first.

Method	What it is	Ballpark $ per acre-foot	kWh/m³	Drought-proof?	Precedent
Stop evaporation	Floating solar and canal lining; the big lakes lose a million acre-feet a year to the sun.	$200–400	~0	No: cuts losses, rain still decides	Project Nexus, California
Pay farms to pause	Paid fallowing and crop shifts; alfalfa is about a third of basin farm use.	$300–500	~0	No: demand response, not supply	Palo Verde fallowing
Reuse wastewater	Treated city and industrial water for farms, cooling and industry.	$600–1,200	0.3–0.6	Partly: sewage flows in a drought	Israel: ~85% reused
Treat farm drainage	Salty irrigation return flows, treated and used again.	$500–1,500	~1	Partly: drainage shrinks with farm cuts	Yuma, idle since 1992
Brackish aquifers	Inland desalination of salty groundwater.	$700–1,500	0.5–1.5	Yes, while the aquifer lasts	El Paso, Texas
Produced water	Oil and gas water, treated for farms or industry.	high today	1–3	Yes, small volumes	Cawelo, California
Potable reuse	Wastewater purified to drinking standard.	$1,500–2,500	1–1.5	Partly	Orange County; NEWater
Seawater desalination	The only source independent of rain, rivers and aquifers.	$2,000–3,000	3–4	Yes: the firm capacity	Carlsbad; Sorek, Israel

Ballpark ranges, verify before quoting. Saline agriculture, growing on salt water, is the 2050 play: it frees fresh water instead of making it.

To finance the firm rung: pay for readiness, not gallons · add modules on a trigger · if it rains, sell or bank the water

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Impact

Timing, size and a payer, in numbers.

A$3B

paid before first water at Melbourne

five idle years at about A$600M a year

⅔

of the capital not committed when you build in stages

first module on the trigger, the rest later

100 kAF

a year kept in Lake Mead when Yuma runs

about the indoor use of 300,000 households

9 markets

lit by the tool, each with a buyer and a payer

on the default priorities, worldwide

Order-of-magnitude estimates from the Australia cases and Reclamation’s Yuma figures.

Path to implementation

Start at Yuma.

Built 1992, run three times in thirty years. All three uncertainties in one idle plant, and Harmony already pilots there.

2026	Run the tool on the Lower Colorado. Publish the map.	Harmony, with CAP, Metropolitan and SNWA
2026–27	Draft the readiness contract. Price the Ciénega offset.	Reclamation, the three states, IBWC and CILA
2027–29	Pilot one module on a Lake Mead trigger.	Reclamation and Harmony
2030+	Credit the first exchange. Repeat in the next markets the tool lights up.	Basin authorities, development banks

Known unknowns: refurbishment cost in the hundreds of millions, NEPA review, the wetland offset with Mexico.

Next steps and team

With more time, and who we are.

Build the data back end

Pull the indices live: Aqueduct, Water Risk Filter, Water Peace and Security, SDG 6, AQUASTAT, GRACE.

More calibration cases

Cape Town, Barcelona, Chennai, Carlsbad.

Talk to the people at Yuma

Reclamation’s office and Harmony’s pilot team: cost and the wetland offset.

Team Sea Major

Mayank D.

Carnegie Mellon · energy systems · finder, film and deck

Tanvir Kabir

Water diplomacy and sources

Teammate 3

Fill in: name and role

Teammate 4

Fill in: name and role

Confluence, live: kugel-blitz-13.github.io/dispatchable-water Sources: aqueduct.wri.org · riskfilter.org (WWF) · waterpeacesecurity.org · sdg6data.org · fao.org/aquastat · nasagrace.unl.edu · climate-conflict.org · usbr.gov (Yuma)